Diatonic Chord Table

Degree	Ionian	Dorian	Phrygian	Lydian	Mixolydian	Aeolian	Locrian
Interval	1	2	3	4	5	6	7
24	①s2	②s2	—	④s2	⑤s2	⑥s2	—
33	①	②m	③m	④	⑤	⑥m	⑦°
34	⑥m^{2}	⑦°^{2}	①^{2}	②m^{2}	③m^{2}	④^{2}	⑤^{2}
42	①s4	②s4	③s4	—	⑤s4	⑥s4	—
43	④^{1}	⑤^{1}	⑥m^{1}	⑦°^{1}	①^{1}	②m^{1}	③m^{1}
44	—	②q	③q	—	⑤q	⑥q	⑦q
233	②m7^{1}	③m7^{1}	④Δ^{1}	⑤7^{1}	⑥m7^{1}	⑦ø7^{1}	①Δ^{1}
323	④Δ^{2}	⑤7^{2}	⑥m7^{2}	⑦ø7^{2}	①Δ^{2}	②m7^{2}	③m7^{2}
332	①6	②m6	①Δ^{3}	④6	⑤6	④Δ^{3}	⑤7^{3}
333	①Δ	②m7	③m7	④Δ	⑤7	⑥m7	⑦ø7
334	①+8	②m+8	③m+8	④+8	⑤+8	⑥m+8	⑦+8
433	①s4+8	②s4+8	—	—	⑤s4+8	—	—
434	④6^{1}	⑤6^{1}	④Δ^{3}	⑤7^{3}	①6^{1}	②m6^{1}	①Δ^{3}
444	—	②q7	③q7	—	—	⑥q7	⑦q7

Key:

no suffix = major (①④⑤) | **m** = minor (②m ③m ⑥m) | **°** = diminished | **ø7** = half-diminished
Δ = major 7th | **7** = dominant 7th | **6 / m6** = major / minor 6th | **q** = quartal
s2 / s4 = sus2 / sus4 | **+8** = root doubled 8ve | **s4+8** = sus4 w/ doubled root | ^{1,2,3} = inversion

Finger Patterns

Pick any pattern for each hand; drill the combination, then swap or advance.

4-fng sequences (8)	1234 2341 3412 4123 4321 3214 2143 1432
3-fng sequences (6)	123 231 312 321 213 132
4-fng split pairs (10)	12|34 23|41 34|12 41|23 13|24 24|13 14|23 23|14 31|42 42|31
3-fng split pairs (6)	13|2 21|3 32|1 1|23 2|31 3|12
2-fng pairs (12)	12 21 13 31 14 41 23 32 24 42 34 43

3

Drill Types

- **Unison** — both hands play the same pattern.
- **Mirror** — one hand forward, other reverse.
- **Contrary** — different patterns per hand; alternate anchor.
- **Cross-size** — 4-fng one hand, 3-fng or 2-fng other.
- **Independent rhythm** — slow anchor + fast cycle; swap.

Hand Technique

- Play with **pads of fingertips**; fingers curved.
- **Thumb** stays above the fingers.
- Wrist **level** with forearm.
- **Close fingers into palm** after plucking.
- Relax between strokes.

Harp Techniques

- **Damping (étouffé)**: closed fingers mute strings between chords.
- **Rolled chord**: fingers 4→1 fast within one beat.
- **Glissando**: set levers to mode, run finger across.
- **Bisbigliando**: soft hand-tremolo on same strings.
- **Harmonic**: palm at midpoint, thumb plucks; 8ve up, half vol.

Practice Progression

- Start **slow**: metronome ≈ 40–60 bpm.
- **4 reps** per pattern before moving on.
- Increase by **4–8 bpm** only when clean.
- Hands **separately first**, then combine.
- **5–10 min daily** beats long rare sessions.

Notation: digits 1–4 are thumb–index–middle–ring. A split pair 12|34 means fingers 1 & 2 strike together, then 3 & 4; 1|23 means 1 alone, then 2 & 3. Sequences play each finger in order.

Harp Cascade Reference

Eb major (E A B strings) · 1=E 2=F 3=G 4=A 5=B 6=C 7=D

↑ asc ↓ desc Addr=(deg)pattern LH/RH alternate, each hand starts one string above the other

The Open Cascade

triads, full range

H		Str	Addr	Chord
LH	↑	E2	(1)33	I
RH	↑	C2	(6)33	vi
LH	↑	A3	(4)33	IV
RH	↑	F4	(2)33	ii
LH	↑	D4	(7)33	vii°
RH	↑	B5	(5)33	V
LH	↓	G5	(3)33	iii
RH	↓	B4	(5)33	V
LH	↓	D3	(7)33	vii°
RH	↓	F3	(2)33	ii

The Tilted Cascade

second inversions, 34

H		Str	Addr	Chord
LH	↑	E2	(1)34	vi^{2}
RH	↑	D2	(7)34	V^{2}
LH	↑	C3	(6)34	IV^{2}
RH	↑	B4	(5)34	iii^{2}
LH	↑	A5	(4)34	ii^{2}
RH	↓	E5	(1)34	vi^{2}
LH	↓	F4	(2)34	vii°^{2}
RH	↓	G3	(3)34	I^{2}
LH	↓	A2	(4)34	ii^{2}

The Grounded Cascade

first inversions, 43

H		Str	Addr	Chord
LH	↑	E2	(1)43	I^{1}
RH	↑	D2	(7)43	vii°^{1}
LH	↑	C3	(6)43	vi^{1}
RH	↑	B4	(5)43	V^{1}
LH	↑	A5	(4)43	IV^{1}
RH	↓	F5	(2)43	ii^{1}
LH	↓	G4	(3)43	iii^{1}
RH	↓	A3	(4)43	IV^{1}
LH	↓	B2	(5)43	V^{1}

The Suspended Cascade

sus2 up, sus4 down

H		Str	Addr	Chord
LH	↑	E2	(1)24	Is2
RH	↑	C2	(6)24	vis2
LH	↑	A3	(4)24	IVs2
RH	↑	F4	(2)24	iis2
LH	↑	B4	(5)24	Vs2
RH	↓	E5	(1)42	Is4
LH	↓	G4	(3)42	iiis4
RH	↓	B3	(5)42	Vs4
LH	↓	C2	(6)42	vis4

The Quartal Cascade

open fourths, 44

H		Str	Addr	Chord
LH	↑	F2	(2)44	iiq
RH	↑	B3	(5)44	Vq
LH	↑	G4	(3)44	iiiq
RH	↑	B5	(5)44	Vq
LH	↓	D4	(7)44	viiq
RH	↓	D3	(7)44	viiq
LH	↓	D2	(7)44	viiq

The Cathedral Cascade

sus4+oct up, sus4 down

H		Str	Addr	Chord
LH	↑	E2	(1)433	Is4+8
RH	↑	F3	(2)433	iis4+8
LH	↑	B4	(5)433	Vs4+8
RH	↓	E5	(1)42	Is4
LH	↓	F4	(2)42	iis4
RH	↓	F3	(2)42	iis4
LH	↓	F2	(2)42	iis4

The Twilight Cascade

7ths root position, 333

H		Str	Addr	Chord
LH	↑	E2	(1)333	IΔ
RH	↑	E3	(1)333	IΔ
LH	↑	E4	(1)333	IΔ
RH	↑	E5	(1)333	IΔ
LH	↓	D4	(7)333	vii∅7
RH	↓	D3	(7)333	vii∅7
LH	↓	D2	(7)333	vii∅7
RH	↓	G2	(3)333	iiim7

The Dusky Cascade

7ths first inversion, 233

H		Str	Addr	Chord
LH	↑	E2	(1)233	IΔ^{1}
RH	↑	D2	(7)233	vii∅7^{1}
LH	↑	C3	(6)233	vim7^{1}
RH	↑	B4	(5)233	V7^{1}
LH	↑	A5	(4)233	IVΔ^{1}
RH	↓	F5	(2)233	iim7^{1}
LH	↓	G4	(3)233	iiim7^{1}
RH	↓	A3	(4)233	IVΔ^{1}
LH	↓	A2	(4)233	IVΔ^{1}

The Arching Cascade

7ths second inversion, 323

H		Str	Addr	Chord
LH	↑	E2	(1)323	IΔ^{2}
RH	↑	D2	(7)323	vii∅7^{2}
LH	↑	C3	(6)323	vim7^{2}
RH	↑	B4	(5)323	V7^{2}
LH	↑	A5	(4)323	IVΔ^{2}
RH	↓	E5	(1)323	IΔ^{2}
LH	↓	F4	(2)323	iim7^{2}
RH	↓	G3	(3)323	iiim7^{2}
LH	↓	G2	(3)323	iiim7^{2}

The Velvet Cascade

6th chords, 332

H		Str	Addr	Chord
LH	↑	E2	(1)332	I6
RH	↑	D2	(7)332	V7^{3}
LH	↑	C3	(6)332	IVΔ^{3}
RH	↑	B4	(5)332	V6
LH	↑	A5	(4)332	IV6
RH	↓	E5	(1)332	I6
LH	↓	F4	(2)332	iim6
RH	↓	G3	(3)332	IΔ^{3}
LH	↓	A2	(4)332	IV6

The Burnished Cascade
6th inversions, 434

H		Str	Addr	Chord
LH	↑	E2	①434	I6[1]
RH	↑	G3	③434	IVΔ^3
LH	↑	B4	⑤434	V6[1]
RH	↑	A5	④434	IV6[1]
LH	↓	C4	⑥434	IΔ^3
RH	↓	C3	⑥434	IΔ^3
LH	↓	C2	⑥434	IΔ^3

The Thundering Cascade
octave doublings, 334

H		Str	Addr	Chord
LH	↑	E2	①334	I+8
RH	↑	F3	②334	ii+8
LH	↑	G4	③334	iii+8
RH	↑	A5	④334	IV+8
LH	↓	D4	⑦334	vii+8
RH	↓	D3	⑦334	vii+8
LH	↓	D2	⑦334	vii+8
RH	↓	F2	②334	ii+8

The Floating Cascade
quartal 7ths, 444

H		Str	Addr	Chord
LH	↑	F2	②444	iiq7
RH	↑	F3	②444	iiq7
LH	↑	G4	③444	iiiq7
RH	↑	G5	③444	iiiq7
LH	↓	D4	⑦444	viiq7
RH	↓	D3	⑦444	viiq7
LH	↓	D2	⑦444	viiq7

The Stormy Cascade
triads→7ths, dramatic

H		Str	Addr	Chord
LH	↑	E2	①33	I
RH	↑	C2	⑥33	vi
LH	↑	A3	④333	IVΔ
RH	↑	A4	④333	IVΔ
LH	↑	A5	④323	IVΔ^2
RH	↓	B5	⑤323	V7[2]
LH	↓	C4	⑥333	vim7
RH	↓	C3	⑥333	vim7
LH	↓	B2	⑤33	V

The Moonrise Cascade
sus→7th→triad

H		Str	Addr	Chord
LH	↑	F2	②42	iis4
RH	↑	C2	⑥24	vis2
LH	↑	A3	④233	IVΔ^1
RH	↑	G4	③233	iiim7[1]
LH	↑	F5	②233	iim7[1]
RH	↓	B5	⑤42	Vs4
LH	↓	D4	⑦43	vii$^{\circ 1}$
RH	↓	E4	①43	I[1]
LH	↓	F3	②43	ii[1]

The Fanfare Cascade
sus4+8 to triads, heroic

H		Str	Addr	Chord
LH	↑	E2	①433	Is4+8
RH	↑	F3	②433	iis4+8
LH	↑	B4	⑤433	Vs4+8
RH	↑	C5	⑥33	vi
LH	↓	A5	④43	IV[1]
RH	↓	B4	⑤43	V[1]
LH	↓	C3	⑥43	vi[1]
RH	↓	D2	⑦43	vii$^{\circ 1}$

The Impressionist Cascade
F column, all 7th patterns

H		Str	Addr	Chord
LH	↑	F2	②44	iiq
RH	↑	F3	②333	iim7
LH	↑	F4	②323	iim7[2]
RH	↑	F5	②233	iim7[1]
LH	↓	C5	⑥332	IVΔ^3
RH	↓	D4	⑦332	V7[3]
LH	↓	E4	①332	I6
RH	↓	E3	①332	I6

The Wistful Cascade
vi family, minor feel

H		Str	Addr	Chord
LH	↑	C2	⑥333	vim7
RH	↑	C3	⑥333	vim7
LH	↑	C4	⑥333	vim7
RH	↑	D4	⑦334	vii+8
LH	↑	D3	⑦334	vii+8
LH	↓	D3	⑦334	vii+8
RH	↓	D2	⑦334	vii+8

The Contemplative Cascade
viq and V7, C column

H		Str	Addr	Chord
LH	↑	C2	⑥44	viq
RH	↑	C3	⑥44	viq
LH	↑	C4	⑥44	viq
RH	↑	B3	⑤333	V7
LH	↑	B4	⑤333	V7
RH	↓	B5	⑤333	V7
LH	↓	C3	⑥444	viq7
RH	↓	C2	⑥444	viq7

The Resolute Cascade
434 all degrees, V+8, vi+8

H		Str	Addr	Chord
LH	↑	E2	①434	I6[1]
RH	↑	F3	②434	iim6[1]
LH	↑	D4	⑦434	V7[3]
RH	↑	B5	⑤334	V+8
LH	↓	C4	⑥334	vi+8
RH	↓	C3	⑥334	vi+8
LH	↓	C2	⑥334	vi+8
RH	↓	F2	②334	ii+8

Composition & Voicing Reference

Cadences

Authentic (perfect)	**V7–I**
Authentic (imperfect)	**V–I**
Half	**I–V *or* ii–V**
Plagal (Amen)	**IV–I**
Deceptive	**V–vi**
Phrygian half	**iv–V**

Common Progressions

Authentic close **I–IV–V–I**
Extended authentic **I–ii–V–I**
Circle of fifths **I–IV–vii°–iii–vi–ii–V–I**
Pop (I–V–vi–IV) **I–V–vi–IV**
Descending bass **I–I^1–IV–IV^1–V**
Deceptive **I–IV–V–vi**
ii–V–I (jazz) **ii**m7–**V**7–**I**Δ
Pachelbel/hymn **I–V–vi–iii–IV–I–IV–V**
Minor (natural) **i–VII–VI–VII–i**
Minor cadential **i–iv–V–i**

Three Types of Chord Motion

Interval buddies (5↔4, 6↔3, 7↔2) share the same feel.

4th	**stronges**	emphasizes target chord
step	**medium**	creates momentum; emphasizes neither
3rd	**weakest**	extension of first chord

- **Strong resolution**: approach home by 4th (V→I)
- **Tonal ambiguity**: approach tonic by 3rd; relative by 4th
- **Flowing phrase**: walk by step, pinning no single chord
- **Chromatic mediant**: 3rd motion between chords from different keys (C→E♭, C→A♭m)
- **Flat-2 (Neapolitan)**: step-down through ♭II brings darkness smoothly

Chord Function

Tonic (T)	**I**, **vi**, **iii**
Subdominant (S)	**IV**, **ii**
Dominant (D)	**V**, **vii°**

Flow: **T**→**S**→**D**→**T**; D resolves to T (or **vi** deceptive).

Non-Chord Tones

Notes outside the underlying chord that decorate voices.

- **Passing tone**: step up/down between two chord tones
- **Neighbor tone**: step away from and back to the same chord tone
- **Suspension (s)**: held over from previous chord, resolves down by step
- **Retardation**: held over, resolves *up* by step
- **Anticipation**: arrives early on a note of the coming chord
- **Appoggiatura**: leapt into, resolves by step (usually down)
- **Escape tone**: left by leap in opposite direction

SATB Voice Ranges

Voice	Low	High
Soprano	C4	G5
Alto	G3	C5
Tenor	C3	G4
Bass	E2	C4

S–A gap: $\leq$ octave A–T gap: $\leq$ octave T–B gap: $\leq$ 12th

Doubling Rules

- Double the **root** in root-position chords
- Double the **bass note** in 1st inversion
- Avoid doubling the **leading tone** (7th degree)
- Avoid doubling the **7th** of a 7th chord
- In **V**7: all 4 notes present—no doubling
- **vii°** triad: double the **3rd** (not root or 5th)

Voice Motion

- **No parallel 5ths / octaves** between any two voices
- **No direct (hidden) 5ths/8ves** in outer voices via similar motion
- **No voice crossing** or **overlap**
- **Contrary** between outer voices is always safe
- **Oblique** (one voice stationary) is safe
- Prefer **stepwise motion**; after a leap, resolve by step opposite
- Avoid **augmented** intervals melodically (esp. aug 2nd)

Tendency Tones

- **Leading tone** (7̂): resolves **up** to tonic
- **Chordal 7th**: resolves **down** by step
- **V**7–**I**: 4̂ down to 3̂; 7̂ up to 1̂
- **Tritone** in **V**7: dim 5th resolves inward to 1̂–3̂

Harmonic Rhythm

- **Slow changes** (1 chord/bar): solemn, spacious, hymn-like
- **Medium** (2/bar): balanced; most common in folk/pop
- **Fast** (4/bar): restless; used for drive or climax
- **Accelerando**: speed up chord changes into cadences for tension
- Stronger chords (I, V) on **stronger beats**; passing chords on weak beats

Modes

Mode	Color	Character
Ionian (1)	major	bright, stable
Dorian (2)	minor	jazzy, folk
Phrygian (3)	minor	Spanish, dark
Lydian (4)	major	floating, dreamy
Mixolydian (5)	major	bluesy, rock
Aeolian (6)	minor	natural minor
Locrian (7)	dim	unstable, rare

Borrowed & Secondary Chords

- **Borrowed**: from parallel minor (♭III, ♭VI, ♭VII, iv) add darkness
- **Relative ambiguity**: major & relative minor share notes—emphasize one via cadence direction
- **V/x**: secondary dominant targeting chord *x*, spells with chromatic leading tone to *x*'s root
- Resolve the secondary LT up, then target resolves normally

Melody Shaping

- Each phrase should have **one high point** (climax)
- Balance **leaps** with **stepwise recovery** in the opposite direction
- **Repeat** motives with variation (sequence, inversion) for coherence
- Align **melodic peaks** with **harmonic peaks** (V, tension chords)
- End phrases on **scale degree 1, 3, or 5** for repose; on 2 or 7 for continuation

Practical Tips

- Keep **common tones** in the same voice when possible
- **I**6 weakens a cadence—use for passing, not finals
- **V–vi** (deceptive): double the **3rd** of **vi**
- **5th** is the most expendable tone—omit before the 3rd
- **Open spacing** suits solemn; **close spacing** is bright
- Avoid **unison** between adjacent voices except at cadences

Harmonizing a Melody

For each phrase, split the melody into **chord tones** and **approach tones**:

- **Start** and **end** the phrase on chord tones of their current chord
- **Land** the resolution (final note) cleanly—if that's solid, the path there can be loose
- Middle notes are **passing** (walk by step), **neighbor** (step away + back), or **enclosure** (overshoot by ½ step + resolve)
- Harmonize passing tones with **passing chords**, not the underlying chord itself
- If the melody note is the chord's **root, 3rd, 5th, or 9th**, it harmonizes cleanly with that chord

Passing Chord Tricks

- **Dominant of target**: V7 of whatever's next pulls strongly toward it
- **Chromatic slide**: drop the target chord down a ½ step for the note before (B6 → C6)
- **Diminished 7ths** resolve to almost anything—great for crunching in between
- Use **ii** in place of **V** (same function, softer)
- **Tritone sub**: ♭II7 replaces V7 (same tritone inside)
- **Back-door cadence**: iv→♭VII→I instead of V→I
- If each **inner voice** sounds good alone, the chord quality doesn't matter much
- **No repeated notes** in inner lines—breaks the flow